# Chapter 3, Section 12

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August 11, 2026

**1.** Let $Y$ be a scheme of finite type over an algebraically closed field $k$. Show that the function

$$\varphi(y) = \dim_k(\mathfrak{m}_y/\mathfrak{m}_y^2)$$

is upper semicontinuous on the set of closed points of $Y$.

*Proof.* Apply (12.7.2) to the sheaf of relative differentials $\Omega_{Y/k}$. For each closed point $y \in Y$, we have a natural isomorphism (II, 8.7A)

$$\mathfrak{m}_y/\mathfrak{m}_y^2 \cong \Omega_{Y/k} \otimes k(y).$$

□

**2.** Let $\{X_t\}$ be a family of hypersurfaces of the same degree in $\mathbb{P}^n_k$. Show that for each $i$, the function $h^i(X_t, \mathcal{O}_{X_t})$ is a constant function of $t$.

*Proof.* Hypersurfaces of same degree in projective space $\mathbb{P}^n$ have the same cohomology. For any hypersurface $X \subset \mathbb{P}^n$ of degree $d$, we have an exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n}(-d) \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_X \longrightarrow 0,$$

and $h^i(X, \mathcal{O}_X)$ is independent of the choice of $X$. □

**4.** Let $Y$ be an integral scheme of finite type over an algebraically closed field $k$. Let $f : X \to Y$ be a flat projective morphism whose fibers are all integral schemes. Let $\mathcal{L}, \mathcal{M}$ be invertible sheaves on $X$, and assume for each $y \in Y$ that $\mathcal{L}_y \cong \mathcal{M}_y$ on the fiber $X_y$. Then show that there is an invertible sheaf $\mathcal{N}$ on $Y$ such that $\mathcal{L} \cong \mathcal{M} \otimes f^*\mathcal{N}$.

*Proof.* If $f$ is flat, then $\mathcal{L} \otimes \mathcal{M}^{-1}$ is flat over $Y$. For each $y \in Y$, $(\mathcal{L} \otimes \mathcal{M}^{-1})_y \cong \mathcal{O}_{X_y}$ on the fibre $X_y$, and each $X_y$ is integral and projective over $k(y)$, so we have

$$h^0(X_y, (\mathcal{L} \otimes \mathcal{M}^{-1})_y) = h^0(X_y, \mathcal{O}_{X_y}) = 1.$$

Thus, $f_*(\mathcal{L} \otimes \mathcal{M}^{-1})$ is locally free of rank one on $Y$ (12.9), say $f^*f_*(\mathcal{L} \otimes \mathcal{M}^{-1}) = \mathcal{N}$ for some $\mathcal{N} \in \operatorname{Pic} Y$. Applying $f^*$ to both sides, and using the fact that the natural morphism

$$f^*f_*(\mathcal{L} \otimes \mathcal{M}^{-1}) \to \mathcal{L} \otimes \mathcal{M}^{-1}$$

is actually an isomorphism because $f_*(\mathcal{L} \otimes \mathcal{M}^{-1})$ is locally free of rank one, we obtain our desired result. □

**5.** Let $Y$ be an integral scheme of finite type over an algebraically closed field $k$. Let $\mathscr{E}$ be a locally free sheaf on $Y$, and let $X = \mathbb{P}(\mathcal{E})$. Then show that $\operatorname{Pic} X \cong (\operatorname{Pic} Y) \times \mathbb{Z}$. This strengthens (II, Ex. 7.9).

*Proof.* Let $n$ be the rank of $\mathcal{E}$, $\mathcal{O}(1)$ the canonical invertible sheaf of $X$, and $\pi : X \to Y$ the natural projection morphism. There is an obvious map

$$\begin{aligned} \varphi : (\operatorname{Pic} Y) \times \mathbb{Z} &\to \operatorname{Pic} X \\ (\mathcal{L}, d) &\mapsto \mathcal{O}(d) \otimes \pi^* \mathcal{L}. \end{aligned}$$

Since $\pi$ is a flat projective morphism, we can apply the results of §12. In particular, if $\mathcal{M}$ is an invertible sheaf on $X$, then $\mathcal{M}_y$ is an invertible sheaf on $X_y \cong \mathbb{P}^n_y$ for all $y \in Y$, where $\mathbb{P}^n_y$ is the projective $n$-space over $k(y)$. In particular, we can write $\mathcal{M}_y \cong \mathcal{O}_{\mathbb{P}^n_y}(d_y)$ for some $d_y \in \mathbb{Z}$. We will show $d_y$ is constant for all $y \in Y$. It will follow by (Ex. 12.4) that there exists an invertible sheaf $\mathcal{N}$ on $Y$ such that $\mathcal{M} \cong \mathcal{O}(d) \otimes \pi^* \mathcal{N}$ for some $d \in \mathbb{Z}$, proving surjectivity of $\varphi$.

We proceed by showing the minimum value of $d_y$ is obtained when $y = \xi$ is the generic point of $Y$. Suppose there exists $y' \in Y$ such that $d_{y'} < d_\xi$. After twisting by $\mathcal{O}(-d_{y'})$, we can assume $d_{y'} = 0$. By (12.8), the function

$$h^0(y, \mathcal{M}) = \dim_{k(y)} H^0(\mathbb{P}^n_y, \mathcal{O}_{\mathbb{P}^n_y}(d_y))$$

is upper semicontinuous on $Y$ (12.8), and we have the strict inequality

$$\dim_{k(y')} H^0(\mathbb{P}^n_{y'}, \mathcal{O}_{\mathbb{P}_{k(y)}}) < \dim_{k(\xi)} H^0(\mathbb{P}^n_\xi, \mathcal{O}_{\mathbb{P}^n_\xi}(d_\xi))$$

by the explicit calculations of (5.1). But this implies the set $\{y \in Y | h^0(y, \mathcal{M}) \geqslant h^0(\xi, \mathcal{M})\}$ is a closed set containing $\xi$ and not containing $y'$, a contradiction. Hence, we have $d_\xi \leqslant d_y$ for all $y \in Y$.

Next, we prove the inequality in the opposite direction. By Serre duality (5.1) (7.7), we have

$$\begin{aligned} h^n(y, \mathcal{M}) &= \dim_{k(y)} H^n(\mathbb{P}^n_y, \mathcal{O}_{\mathbb{P}^n_y}(d_y)) \\ &= \dim_{k(y)} H^n(\mathbb{P}^n_y, \mathcal{O}_{\mathbb{P}^n_y}(-d_y - n - 1)) \\ &= h^0(y, \mathcal{O}(-n-1) \otimes \mathcal{M}^{-1}), \end{aligned}$$

and $h^n(y, \mathcal{M})$ is also an upper semicontinuous on $Y$. Suppose there exists $y' \in Y$ such that $d_\xi < d_{y'}$. After twisting by $\mathcal{O}(-d_\xi + n + 1)$, we can assume $d_\xi = n + 1$. In particular $d_{y'}$ is a positive integer. Again, we have the strict inequality

$$\dim_{k(y')} H^0(\mathbb{P}^n_{y'}, \mathcal{O}_{\mathbb{P}^n_{y'}}(-d_{y'} - n - 1)) < \dim_{k(\xi)} H^0(\mathbb{P}^n_\xi, \mathcal{O}_{\mathbb{P}^n_\xi}),$$

arriving at the same contradiction as above. Hence, $d_y$ is constant for all $y \in Y$, and we have shown $\varphi$ is surjective.

It remains to show $\varphi$ is injective. If we have an isomorphism

$$\mathcal{O}(d) \otimes \pi^* \mathcal{L} \cong \mathcal{O}(d') \otimes \pi^* \mathcal{L}',$$

then it is clear by checking on fibres that $d = d'$, so it suffices to check $\pi^* : \operatorname{Pic} Y \to \operatorname{Pic} X$ is injective. Indeed, we have $\pi_* \mathcal{O}_X = \mathcal{O}_Y$ (II, 7.11a), and the projection formula gives

$$\pi_* \pi^* \mathcal{N} \cong \pi_* \mathcal{O}_X \otimes \mathcal{N} \cong \mathcal{N}$$

for all $\mathcal{N} \in \operatorname{Pic} Y$, so if $f^* \mathcal{N} \cong \mathcal{O}_X$, then

$$\mathcal{N} \cong \pi_* \mathcal{O}_X \cong \mathcal{O}_Y.$$

$\square$